

UK Shock Propagation: The Real Bloomberg-Data Methodology

How Revenue and EV/EBITDA multiple shocks are propagated from GICS Sector to GICS Sub-Industry, UK only, using real Bloomberg market data throughout.

Companion to **UK_Revenue_EVEBITDA_Cascade.xlsx** — every figure below is a live cell in that workbook. Supersedes *UK_EVEBITDA_Methodology_Change.pdf*.

The method in one line: the PRA gives a single shock per GICS Sector (11 categories). Each sub-industry's shock is that sector figure multiplied by a ratio built from real, current UK market data — its own market metric relative to its sector's real, value-weighted average — so the weighted average of the outputs reconstructs the PRA's sector number exactly, and the granular differentiation between sub-industries is read off the market, not assumed.

1. EV/EBITDA — the multiple itself is the differentiator

For every sub-industry with a UK-listed constituent, the shock is built entirely from real numbers:

$$Shock_i = Shock_{Sector} \times \frac{M_i}{\bar{M}_{Sector}}$$

M_i = sub-industry i's real current EV/EBITDA multiple • $\bar{M}_{Sector}$ = the sector's real, market-cap-weighted average multiple (both from Bloomberg EQS<GO>, UK universe, 8 Aug 2026)

This is not a proxy standing in for the thing being measured — it is the thing being measured, one GICS level more granular than the PRA published. **125 of 163** UK sub-industries have a usable multiple this way.

1.1 Why the multiple's own level is the right differentiator

Approximate any EBITDA-generating business as a growing perpetuity, so its multiple is the inverse of the gap between the discount rate and the growth rate:

$$M = \frac{1}{r - g} \Rightarrow \frac{dM}{M} = -dr$$

the PERCENTAGE change in a multiple, for a given rise in the discount rate, is proportional to the multiple's own starting level

THE MECHANISM

This scenario widens discount rates and risk premia across the board. A sub-industry priced at 20x compresses roughly twice as hard, in percentage terms, as one at 10x, because more of its value sits in distant cash flows — textbook duration. Using the multiple relative to its sector average as the differentiator isn't a modelling choice layered on top of the data; it falls directly out of differentiating the valuation identity every multiple is built on.

1.2 A data-quality correction, applied at the source

Two sub-industries' current-year multiples were near-zero-EBITDA artifacts (213.01x and 58.62x, driven by a single constituent), which would have implied a compression beyond –100% — not merely large, but physically impossible for a multiple. Both are corrected in the source data (`3_UK_Bloomberg_Data` , column "Multiple Used") to Bloomberg's own forward ("BF") multiple — 9.8x and 16.2x, in line with neighbouring sub-industries — flagged and documented in column I of that sheet. The cascade formulas themselves are untouched; only the two flagged inputs were corrected.

2. Revenue — real UK data sets the weight; here is why that's the right and honest use of it

Revenue works differently, and the difference matters enough to state plainly rather than gloss over.

2.1 What the data can and can't tell us

The PRA's own Revenue figure is not a genuine per-sector calculation the way EV/EBITDA is — it is a three-tier severity label. Proof, not assertion: US Year-1 revenue is `-10.0000%` for five different sectors (Consumer Discretionary, Financial Services, Industrial, Real Estate, Technology) and `-5.0000%` for four others, identical to the last decimal place — five sectors assigned one label, four assigned another, not five independent cross-sectional calculations that happen to converge. There is consequently no finer *revenue-growth-sensitivity* signal hiding beneath the sector number for real UK data to extend, the way there was for EV/EBITDA. An external cyclicity measure (Damodaran unlevered beta, credibility-shrunk) supplies that sensitivity axis — it is the best available answer to "how much more, or less, than the sector average should this sub-industry move," and remains so however the aggregate is weighted.

2.2 What real UK data legitimately fixes: the weight

What real data *does* settle, honestly, is how much each sub-industry should count in the sector average that the sensitivity ratio is measured against — and that has nothing to do with cyclicity. A revenue-shock aggregate is, definitionally, a revenue-weighted average: if a sector's sub-industries are weighted equally regardless of size, a sub-industry with two small companies pulls the sector average exactly as hard as one with forty large ones, which is not how the PRA's own sector figure was constructed (their revenue definition is explicitly a cross-sectional average *of company revenues*). Real UK Revenue T12M (Bloomberg EQS<GO>, GICS Sub-Industry, 8 Aug 2026) replaces the equal-weight default with the true, current revenue base of each sub-industry:

$$Shock_i = Shock_{Sector} \times RelBeta_i$$

RelBeta cascades L2→L3→L4 exactly as before, but every average in the chain is now weighted by real UK Revenue T12M rather than counted equally

135 of 163 UK sub-industries have usable real revenue data this way (4 more show a literal 0.00 in the source and are treated as no-data, not zero: Coal & Consumable Fuels, Drug Retail, Industrial Gases, Motorcycle Manufacturers).

3. Data quality

The UK figures behind both cascades were transcribed from two independent Bloomberg exports — EV/EBITDA and Revenue — captured on different screens. **139 of 139 sub-industry names, and 139 of 139 market-capitalisation figures, matched exactly between the two exports** (0 mismatches), which is the cross-check that gives confidence the transcription is accurate: two independently-read screenshots of the same underlying Bloomberg universe agreeing to the pound on every market cap is not something that happens by chance.

4. Consistency proof — reconciliation against the PRA's own numbers

Because every real-data shock is the sector input times a ratio normalised to that sector's own real, weighted average, the real-weighted mean of the sub-industry outputs must reconstruct the PRA's Level-1 figure exactly — algebra, not a fitted result. Verified on the saved workbook (`6_Control_Check` , Blocks 3 & 4), all eleven sectors, both metrics:

Sector	Revenue L1	Real-weighted recon.	EV/EBITDA L1	Real-weighted recon.
Energy	+8.0%	+8.0000%	+0.0%	+0.0000%
Materials	−5.0%	−5.0000%	−28.0%	−28.0000%
Industrials	−10.0%	−10.0000%	−32.0%	−32.0000%
Consumer Discretionary	−10.0%	−10.0000%	−25.0%	−25.0000%
Consumer Staples	−5.0%	−5.0000%	−25.0%	−25.0000%
Health Care	−5.0%	−5.0000%	−25.0%	−25.0000%
Financials	−10.0%	−10.0000%	−32.0%	−32.0000%
Information Technology	−10.0%	−10.0000%	−36.0%	−36.0000%
Communication Services	−5.0%	−5.0000%	−26.0%	−26.0000%
Utilities	+2.0%	+2.0000%	−1.0%	−1.0000%
Real Estate	−10.0%	−10.0000%	−36.0%	−36.0000%

Variance in the live workbook is floating-point noise ($\sim 1e-16$ to $1e-17$), not a fitted approximation — shown here rounded to 4 decimal places to fit the page. Reconstruction uses real UK weights (revenue T12M for the Revenue block, market cap for the EV/EBITDA block) over the sub-industries with real data only, exactly as the cascade computes them.

5. What this does and does not claim

- **EV/EBITDA** is real Bloomberg data end to end — both the differentiator (the multiple) and the weight (market cap) — for 125 of 163 sub-industries. The other 38 (no UK listing) fall back to the beta-cascade estimate, explicitly flagged, not silently blended.
- **Revenue** is real Bloomberg data for the weight (135 of 163 sub-industries), applied to a beta-based sensitivity ratio — because no real per-sub-industry revenue-growth signal exists to extend; the PRA's own figure doesn't have one either. This is stated as what it is, not overstated as fully real-data-driven.
- Both cascades reconstruct the PRA's published Level-1 numbers exactly, by construction, for the sub-industries they cover.

Sources: Bloomberg EQS<GO>, UK Refined/Investable Universe, GICS Sub-Industry views for EV/EBITDA and Revenue T12M (read 8 Aug 2026). BoE Private Markets SWES variable-path annex, "Sector revenues" and "Equity multiples"

blocks (UK Year-1). Damodaran Industry Betas, NYU Stern (5 Jan 2026), used for Revenue's sensitivity axis only.
Companion file: UK_Revenue_EVEBITDA_Cascade.xlsx.